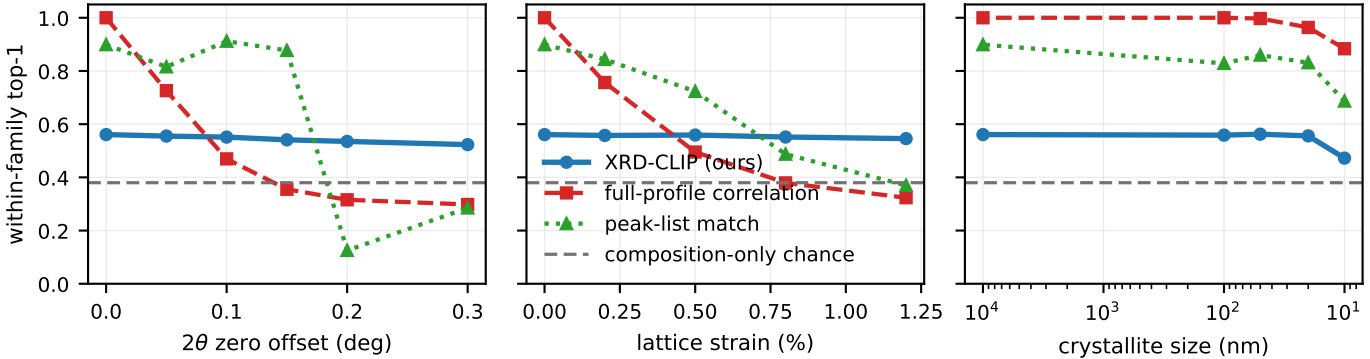
(a) sample displacement / zero error
(b) thermal / compositional strain
(c) Scherrer broadening (limitation)
within-family top-1
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.1
0.2
0.3
2θ zero offset (deg)
0.00
0.25
0.50
0.75
1.00
1.25
lattice strain (%)
XRD-CLIP (ours)
full-profile correlation
peak-list match
composition-only chance
10^4
10^3
10^2
10^1
crystallite size (nm)